

Exam: Statistical Data Visualization

Practice Exam with Solutions | Basic Level

1. **[Application] (4 Pts.)** Two groups ($n=50$), metric DV. Best plot? Was does bar+error bar miss?
2. **[Open question] (3 Pts.)** Scatterplot 500 points overplotting. 3 methods. Why no 3D bars?
3. **[Open question] (2 Pts.)** Forest plot purpose? What does the diamond show?
4. **[Open question] (3 Pts.)** Four Tufte principles?
5. **[Open question] (2 Pts.)** Which plot shows distribution + group comparison + individual points?

SOLUTIONS

1. Raincloud plot. Bar+error bar hides distribution shape, outliers, n, median/quantiles.
2. 1) Alpha transparency. 2) Jitter. 3) 2D histogram. 3D distorts proportions.
3. Meta-analysis: study points + CI. Diamond = pooled effect with CI.
4. 1) Maximize data-ink ratio. 2) No chartjunk. 3) Small multiples. 4) Graphical integrity.
5. Raincloud plot (violin + boxplot + stripplot) or beeswarm + boxplot.